AI LAB 4 CALCULATIONS

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Step 3:

P(none, no, on time, attend)

= P(none) × P(no | none) × P(on time | none, no) × P(attend | on time)

= 0.7 × 0.6 × 0.9 × 0.9 = 0.3402

Step 4:

P(heavy, yes, delayed, miss)

= P(heavy) × P(yes | heavy) × P(delayed | heavy, yes) × P(miss | delayed)

= 0.1 × 0.1 × 0.6 × 0.4 = 0.0024

Step 6:

X = Rain, e = train delayed, hidden variable = Maintenance

P(none, delayed)

= 0.7 × (0.4 × 0.2 + 0.6 × 0.1) = 0.7 × 0.14 = 0.098

P(light, delayed)

= 0.2 × (0.2 × 0.4 + 0.8 × 0.3)

= 0.2 × 0.32

= 0.064

P(heavy, delayed)

= 0.1 × (0.1 × 0.6 + 0.9 × 0.5)

= 0.1 × 0.51

= 0.051

Sum = 0.098 + 0.064 + 0.051 = 0.213

Normalize: P(rain = none | train = delayed) = 0.098 / 0.213 = 0.4601